Jairam Alluri

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I'm a Machine Learning Engineer with **3+ years** of experience building ML systems across computer vision, NLP, and LLMs. I hold a **USPTO patent approved for filing** and have driven ML work central to a **$10M business line**, delivering impact on platforms serving **600+ TB** of data and hundreds of thousands of users.

EDUCATION

University of Texas — **Aug 2022 – May 2024**
M.S. in Business Analytics — Data Science Cohort — *Richardson, TX*

GPA: 3.91/4.0 | Ranked #1 in AI/ML and Statistics courses | Coursework: Machine Learning, Deep Learning, Statistical Modeling, Natural Language Processing

SKILLS

Languages: Python, SQL

ML & AI: PyTorch, TensorFlow, Scikit-learn, Transformers (BERT, ViT, LLaMA, GPT-2), YOLO, OpenCV, Computer Vision, NLP, Edge ML, Time Series

LLM & GenAI: LangChain, Retrieval-Augmented Generation (RAG), RAG Fusion, Multi-Query Retrieval, Prompt Engineering, LLM-as-judge, DeepEval, BM25, Embeddings, Vector Databases (ChromaDB, FAISS), HuggingFace

MLOps & Cloud: AWS (EC2, Bedrock, S3), Azure (Auto-scaling, Spot Instances), Docker, Flask, REST API, MLflow, CI/CD, GitHub Actions

EXPERIENCE

Honeywell — **May 2023 – Present**
Machine Learning Engineer — *Richardson, TX*

- I owned KPI generation across a **600+ TB dataset** powering ML training on a building automation platform serving **hundreds of thousands of users**, improving model performance by **70% on under-represented classes**.
- I drove ML development for a 3D computer vision system using PyTorch and OpenCV with a custom Vision Transformer (ViT) and specialized positional encodings; my ML work was central to a **$10M business line reaching its target goals**, with the data-preprocessing technique **approved for filing with the USPTO**.
- I improved data processing **10×** within the vision system via parallel execution while **simultaneously cutting Azure cloud costs by 80%** through workload-specific instance selection and spot-instance migration.
- I built a radar-based ML system for real-time sensing, maintained as an internal Honeywell **trade secret**.
- I developed and deployed TensorFlow edge ML models for real-time gas concentration prediction on resource-constrained embedded hardware with minimal CPU overhead.

PROJECTS

Academic Knowledge Assistant — Hybrid RAG Pipeline | *LangChain, ChromaDB, BM25, AWS Bedrock, Flask, Docker* — Live Demo · GitHub

- I combined ChromaDB dense vector search with BM25 sparse retrieval, fused via Reciprocal Rank Fusion (RRF), **improving recall ~30%** on technical queries; I added Multi-Query Retrieval for complex questions, returning grounded answers with source citations.

GPT-2 Language Model Built from Scratch | *PyTorch, Custom Tokenizer* — Medium

- I implemented the full GPT-2 decoder-only architecture from scratch (Masked Multi-Head Self-Attention, Sinusoidal Positional Encodings, Layer Normalization, Residual Connections) with **no external transformer libraries**; I trained it on a **1M+ token corpus**.

Deep Learning Autocorrect System | *PyTorch, Seq2Seq, FastText, 1D-CNN, Flask* — Medium

- I built a Seq2Seq Encoder-Decoder model with Attention and 1D-CNN layers for real-time sentence-level correction; I deployed it as a Flask app with **sub-100ms end-to-end inference**.

LEADERSHIP & AWARDS

- **Patent (Approved for Filing, USPTO):** Novel AI-based data-preprocessing technique for healthcare testing, Honeywell International, 2025. [Announcement]
- **Dean's Excellence New Student Cohort Scholarship**, University of Texas, 2022 — awarded for exceptional academic performance at entry.